sumptions, the steady-state turnover rate of the exchanger can be derived from equation 65a, and the corresponding fluxes of substrates A, B, and C are obtained through equations 65b, 65c, and 65d, respectively.

A general simplified antiporter transport with three substrates

$$\boxed{\mathrm{aA_M + bB_N + cC_N \rightleftharpoons aA_N + bB_M + cC_M}}$$

Net Transport Flux from M to N

$$\boxed{J^{M,N(net)}_{antiporter} = P_{antiporter}\frac{(\alpha^M)^a(\beta^N)^b(\gamma^N)^c - (\alpha^N)^a(\beta^M)^b(\gamma^M)^c}{R'_M R'_{NN} + R'_N R'_{MM}}} \tag{65a}$$

$$\boxed{J^{M,N(net)}_{A,antiporter} = aJ^{M,N(net)}_{antiporter}} \tag{65b}$$

$$\boxed{J^{M,N(net)}_{B,antiporter} = bJ^{M,N}_{antiporter}} \tag{65c}$$

$$\boxed{J^{M,N(net)}_{C,antiporter} = cJ^{M,N(net)}_{antiporter}} \tag{65d}$$

where $P_{antiporter} = [E]_t g_{antiporter}$ describes the membrane permeability to the antiporter, and the thermodynamic parameters and resistances are defined as:

$$\alpha^M = \frac{[A]_M}{K_A},\ \beta^M = \frac{[B]_M}{K_B},\ \gamma^M = \frac{[C]_M}{K_C} \qquad \alpha^N = \frac{[A]_N}{K_A},\ \beta^N = \frac{[B]_N}{K_B},\ \gamma^N = \frac{[C]_N}{K_C}$$

$$R'_M = (1+\alpha^M)^a(1+\beta^M)^b(1+\gamma^M)^c \qquad R'_N = (1+\alpha^N)^a(1+\beta^N)^b(1+\gamma^N)^c$$

$$R'_{MM} = (a\alpha^M + b\beta^M + c\gamma^M) \qquad R'_{NN} = (a\alpha^N + b\beta^N + c\gamma^N)$$

Figure 22: Antiporter Simplified Model. The following assumptions are used: (1) the rapid binding and unbind assumptions, (2) the quasi-steady-state assumption, (3) similarity assumption for all rate constants (4) antiporter has a single site which bound the transported solute (5) the transported solute competed for the same binding sites (6) finally, the empty antiporter does not cross the membrane.

6. Discussion and Conclusion

Multiscale mathematical modeling of biological transport mechanisms in the human body has advanced significantly over the last few decades. Simultaneous advances in experimental techniques and computational power have produced models with much greater predictive power, complexity, and usefulness; these models can be used to achieve a more thorough understanding of membrane physiology and disorders. These advances have also led to new techniques in the pre-